

# **COMPARATIVE ANALYSIS OF DEEP LEARNING ARCHITECTURES FOR SMARTPHONE-BASED HUMAN ACTIVITY RECOGNITION**

**Synopsis on  
Deep Learning Project  
[ICT-4442]**

Submitted By

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## I. BRIEF DESCRIPTION OF THE PROJECT

Human Activity Recognition (HAR) is the automated identification of physical activities from sensor observations. This project proposes a comparative study of three deep learning architecture families—Multilayer Perceptron (MLP), one-dimensional Convolutional Neural Network (1D-CNN), and Long Short-Term Memory (LSTM)—for recognizing everyday human activities from inertial sensor data collected by a smartphone. The selected UCI Human Activity Recognition Using Smartphones dataset contains recordings from 30 volunteers performing six activities: walking, walking upstairs, walking downstairs, sitting, standing, and laying. The smartphone was worn at the waist and its embedded accelerometer and gyroscope recorded motion at 50 Hz [1].

The study will focus on the raw multichannel inertial signals rather than relying only on the dataset's pre-engineered 561-feature representation. Each observation window will therefore be represented as a fixed-length sequence of sensor measurements. The same subject-wise training and test partition will be maintained for all models, while a validation subset will be formed only from the training subjects. This provides a common experimental setting for comparing the architectures using accuracy, precision, recall, F1-score, confusion matrices, and computational considerations. The MLP will provide a simple fully connected baseline after flattening each window; the 1D-CNN will learn local temporal patterns in the sensor sequence; and the LSTM will model temporal dependencies across the sequence. The comparison will investigate not only which model performs best, but also why their performance differs across static and dynamic activities.

HAR has practical relevance to fitness tracking, context-aware mobile applications, ambient assisted living, rehabilitation, and remote activity monitoring. Sensor-based recognition is also attractive where camera-based monitoring would create privacy or deployment constraints [2], [3]. The project therefore provides a compact but meaningful deep learning study with a clear path toward wearable and mobile inference.

## II. DATASET(S) IDENTIFIED

### A. UCI Human Activity Recognition Using Smartphones

The project will use the UCI HAR dataset (UCI Machine Learning Repository, dataset 240). It contains 10,299 labeled instances derived from 30 volunteers aged 19–48. The volunteers performed six activities while carrying a Samsung Galaxy S II at the waist. Triaxial total/body acceleration and triaxial angular velocity were collected at 50 Hz. The repository provides both raw inertial signals and a 561-feature time/frequency representation [1]. The original subject-wise split uses approximately 70% of the volunteers for training and 30% for testing; this split will be retained to reduce subject leakage. A validation set will be created from the training subjects only. The raw input will use nine sensor channels and fixed 128-sample windows, allowing all three proposed architectures to receive the same underlying observations.

## III. PROPOSED MODELS FOR COMPARISON

### A. Multilayer Perceptron (MLP)

A compact fully connected neural network will operate on flattened sensor windows. It serves as an interpretable deep-learning baseline and tests how well dense layers can classify activities without explicit temporal or local pattern modeling.

### B. One-Dimensional Convolutional Neural Network (1D-CNN)

A 1D-CNN will apply temporal convolutions over the sensor channels to learn local motion patterns. CNN-based approaches have previously shown strong suitability for multivariate smartphone sensor time series [4].

### C. Long Short-Term Memory (LSTM)

An LSTM network will process the sensor window sequentially and learn temporal dependencies between measurements. LSTM-based models are particularly relevant because human activities are sequences of movements rather than isolated sensor values [5].

TABLE I. PROPOSED MODELS AND MODEL-TO-MEMBER ASSIGNMENT

| Member             | Assigned Model / Task | Primary Responsibility                                             |
|--------------------|-----------------------|--------------------------------------------------------------------|
| Shreyas Bibhuty    | MLP baseline          | Baseline architecture, preprocessing checks, tuning and evaluation |
| Ansh Aaditya Singh | 1D-CNN                | Temporal convolution model, tuning and error analysis              |
| Aryan Ranpura      | LSTM                  | Sequential model, tuning and error analysis                        |

## IV. EXISTING STATE-OF-THE-ART METHOD(S) RELATED TO THE PROPOSED PROJECT

Existing HAR research has moved from handcrafted statistical and frequency-domain features toward end-to-end deep learning on raw sensor sequences. Zeng et al. demonstrated CNN-based feature learning for mobile-sensor activity recognition, showing that convolution can exploit local dependencies in sensor signals [4]. Ordóñez and Roggen proposed DeepConvLSTM, combining convolutional layers for feature extraction with LSTM layers for temporal modeling and sensor fusion [5]. More recent work has explored hybrid CNN-GRU architectures, attention mechanisms, and other deep architectures for wearable HAR [6], while Zhang et al. provide a broad review of deep-learning approaches, datasets,

deployment considerations, and open challenges in wearable HAR [3]. These studies motivate the proposed comparison: MLP provides a simple baseline, CNN tests local temporal feature extraction, and LSTM tests explicit sequence modeling. Rather than attempting to reproduce a large state-of-the-art architecture, this project will use these representative families to study the trade-off between accuracy, model complexity, and temporal modeling capability on a common benchmark.

## V. OBJECTIVES

- To compare the effectiveness of MLP, 1D-CNN, and LSTM architectures for six-class human activity recognition from smartphone inertial sensor sequences.
- To determine how architecture choice affects classification performance, particularly for activities with similar motion patterns such as sitting versus standing and walking upstairs versus downstairs.
- To assess the practical trade-off between predictive performance and model complexity for potential smartphone or wearable deployment.

## VI. IMPORTANCE / MOTIVATION OF THE PROPOSED PROJECT

Human activity recognition is a practical sensing problem underlying fitness tracking, rehabilitation support, ambient assisted living, context-aware mobile computing, and activity monitoring. Wearable and smartphone sensors offer a less intrusive alternative to camera-based recognition in settings where privacy, lighting, or physical deployment make vision systems unsuitable [2], [3]. Although the selected dataset is a standard benchmark, the project is designed around a real deployment question: how should a deep-learning architecture be selected when sensor data are sequential, resource-constrained, and generated continuously? Comparing three distinct architecture families under identical data splits and metrics makes the study both educationally suitable for a beginner-level deep-learning team and relevant to real-world edge/wearable AI. The resulting analysis can also establish a baseline for future extensions such as on-device inference, personalization to unseen users, additional activities, and lightweight models for smartphones.

## VII. PRELIMINARY REFERENCE LIST

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- [6] M. A. K. Kushwaha, S. Kar, and S. Dwivedi, "Deep learning based human activity recognition (HAR) using wearable sensor data," *International Journal of Information Management Data Insights*, vol. 1, no. 2, p. 100046, 2021, doi: 10.1016/j.jjime.2021.100046.

## VIII. WORK PLAN

TABLE II. WORK PLAN AND EXPECTED COMPLETION DATES

| Sl. No. | Component / Work Element / Milestone                                                                                           | Expected Completion |
|---------|--------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 1       | Finalize problem statement, dataset, literature search and synopsis                                                            | 25 Aug 2026         |
| 2       | Dataset acquisition, subject-wise split verification and preprocessing pipeline                                                | 31 Aug 2026         |
| 3       | MLP baseline implementation and preliminary evaluation                                                                         | 08 Sep 2026         |
| 4       | 1D-CNN implementation and preliminary evaluation                                                                               | 12 Sep 2026         |
| 5       | LSTM implementation and preliminary evaluation                                                                                 | 16 Sep 2026         |
| 6       | Hyperparameter tuning, debugging and individual model error analysis                                                           | 20 Sep 2026         |
| 7       | Common evaluation using accuracy, precision, recall, F1-score and confusion matrices; comparative analysis of all three models | 25 Sep 2026         |
| 8       | Final tuning, extended experiments, report and presentation preparation                                                        | 31 Oct 2026         |